

FINAL PROJECT REPORT

STAT - 315

CUSTOMER CHURNS

Using Logistic Regression Analysis

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Purpose of Data Modelling:

- Understanding customer churn is pivotal for banks due to the high costs associated with acquiring new customers versus retaining existing ones. This study analyses a dataset from a bank to identify why customers leave, focusing on variables such as Credit Score, Age, Account Balance, and Engagement. By examining these factors, the research aims to uncover the dynamics behind churn, emphasizing the financial and demographic attributes that influence a customer's decision to exit.
- The motivation behind this research is the significant impact churn has on a bank's profitability and growth. In the competitive banking industry, retaining customers is not only economically sensible but also critical for maintaining a stable customer base. The analysis seeks to provide banks with insights into developing targeted strategies for customer retention, enhancing service quality, and tailoring products to meet customer needs effectively.
- The goal is to offer banks actionable strategies for reducing churn through an in-depth understanding of customer behaviour and preferences. By focusing on the key determinants of churn identified in this study, banks can implement loyalty programs, improve customer satisfaction, and ultimately, foster a more loyal customer base, securing their market position in the long run.

Data Set Overview:

Data set has a sample size of 1000 observations. Various qualitative and quantitative variables.

CustomerId—contains random values and has no effect on customer leaving the bank.

Surname—the surname of a customer has no impact on their decision to leave the bank.

CreditScore—can have an effect on customer churn, since a customer with a higher credit score is less likely to leave the bank.

Geography—a customer's location can affect their decision to leave the bank.

Gender—it's interesting to explore whether gender plays a role in a customer leaving the bank.

Age—this is certainly relevant, since older customers are less likely to leave their bank than younger ones.

Tenure—refers to the number of years that the customer has been a client of the bank. Normally, older clients are more loyal and less likely to leave a bank.

Balance—also a very good indicator of customer churn, as people with a higher balance in their accounts are less likely to leave the bank compared to those with lower balances.

NumOfProducts—refers to the number of products that a customer has purchased through the bank.

HasCrCard—denotes whether or not a customer has a credit card. This column is also relevant, since people with a credit card are less likely to leave the bank.

IsActiveMember—active customers are less likely to leave the bank.

EstimatedSalary—as with balance, people with lower salaries are more likely to leave the bank compared to those with higher salaries.

Exited—whether or not the customer left the bank.

Complain—customer has complaint or not.

Satisfaction Score—Score provided by the customer for their complaint resolution.

Card Type—type of card holds by the customer.

Points Earned—the points earned by the customer for using credit card.

How we structured our model:

Step 1: Create a Model using all variables in R.

Step 2: Use Backward Selection to find which is the best model.

Step 3: After consideration, logically come up with more variables that must be included in the model.

Step 4: Try doing Analysis of that model using Plots.

Step 5: After Analysis design a Model that suits your needs.

Model designing:

Model 1: We constructed a model containing all the variables in our data set and named it as Model 1.

Exited ~ CreditScore + Geography + Gender + Age + Balance + NumOfProducts + HasCrCard + IsActiveMember + EstimatedSalary + EstimatedSalary + Complain + Satisfaction_Score + Card.Type + Point.Earned

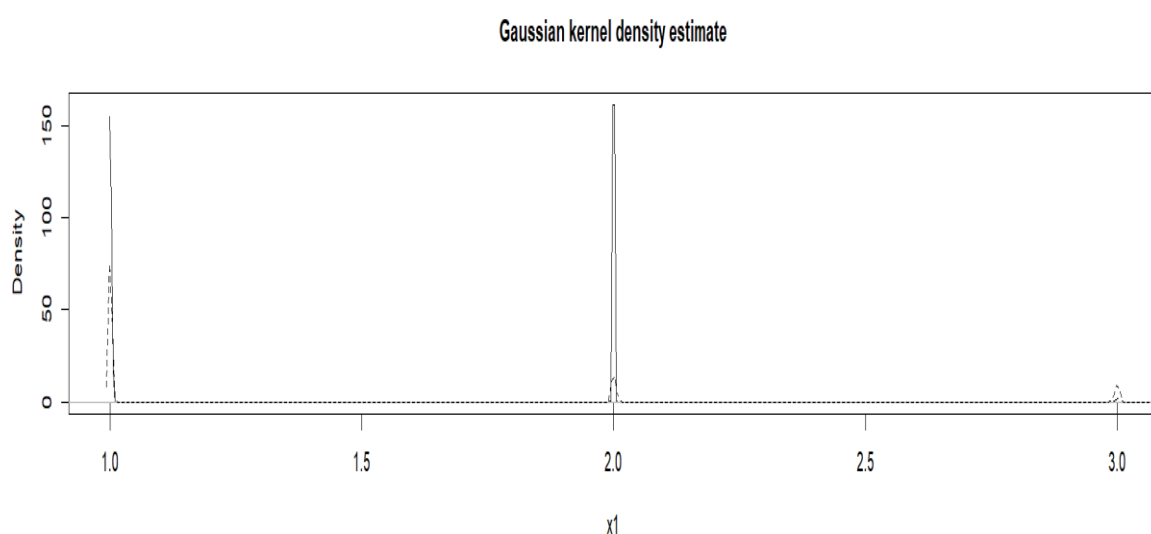
Model 2: Now we will consider some sort of best model selection method, here we applied backward selection, and we got our Model 2:

Exited ~ Age + IsActiveMember + Satisfaction_Score + Complain + Point.Earned

Model 3: After carefully consideration, we use some logic and based on our R&D, we will be adding a few extra variable namely Balance and Numofproducts to our model. This will give us a better idea of understanding customer churning.

Exited ~ Age + IsActiveMember + Satisfaction_Score + Complain + Point.Earned + Balance + NumOfProducts

Model 4: After considering Gaussian Plot we decided to take log transformation.



Exited ~ Age + IsActiveMember + Satisfaction_Score + Complain + Point.Earned
+ log(NumOfProducts) + NumOfProducts

Significance Level = 10%

Reference Category for Satisfaction Score is SatisfactionScore1

Best Model:

$$\text{Log}\frac{(\hat{\theta})}{(1-\hat{\theta})} = -18.8207 + 0.0849\text{Age} - 1.5667\text{IsActiveMember} + \\ 0.6873\text{SatisfactionScore2} - 1.8683\text{SatisfactionScore3} + \\ 0.5563\text{SatisfactionScore4} - 1.4080\text{SatisfactionScore5} + 14.4887\text{Complain} - \\ 0.0026\text{PointEarned} - 17.4843\ln(\text{NumOfProduct}) + 10.3263\text{NumOfProduct}$$

We did our analysis at 10% significance level.

Here reference category for satisfaction Score is SatisfactionScore1.

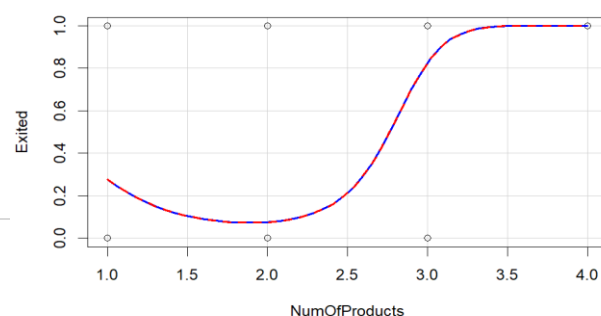
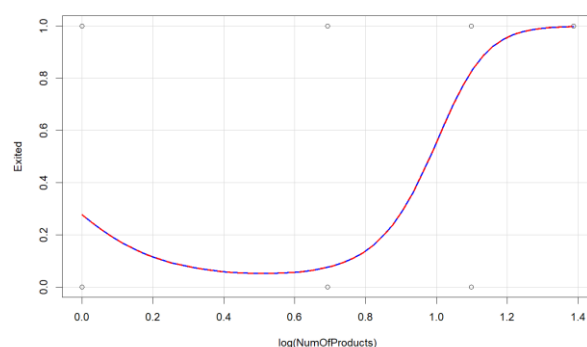
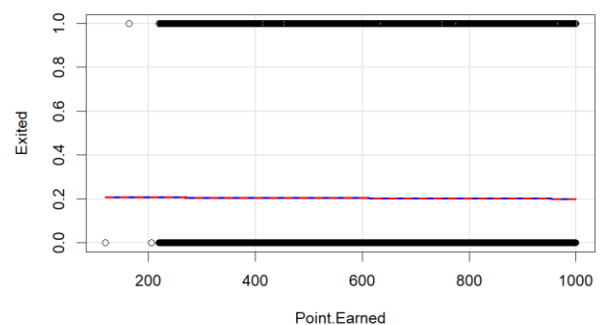
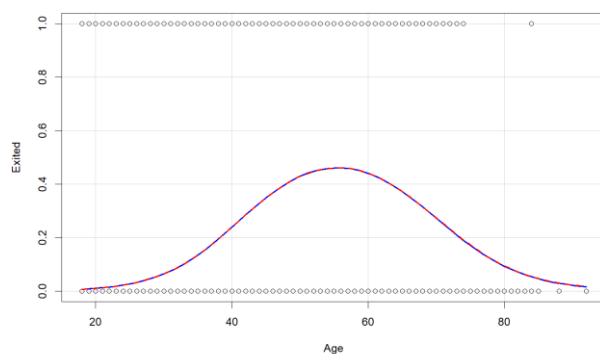
Interpretations of the coefficients:

- Constant Term ($e^{(-18.8207)}$): The baseline odds of a customer churning, when all other variables are zero, are approximately $e^{-18.8207}$. This is a very small odds, indicating that churning is extremely unlikely in the absence of other contributing factors.
- Age ($e^{(0.0849)}$): Each additional year of age multiplies the odds of churning by $e^{0.0849} \approx 1.089$. This means that the odds of churning increase by about 8.9% with each additional year of age.
- IsActiveMember ($e^{(-1.5667)}$): Being an active member divides the odds of churning by $e^{1.5667} \approx 4.79$. Therefore, active members are about 79% less likely to churn compared to non-active members.
- Satisfaction Scores:
 - SatisfactionScore2 ($e^{(0.6873)}$): Customers with a satisfaction score of 2 have their odds of churning multiplied by $e^{0.6873} \approx 1.99$, almost doubling the odds compared to the baseline score.
 - SatisfactionScore3 ($e^{(-1.8683)}$): Customers with a satisfaction score of 3 have their odds of churning divided by $e^{1.8683} \approx 6.47$, significantly lowering the risk of churning.

- SatisfactionScore4 ($e^{(0.5563)}$): These customers have their odds of churning multiplied by $e^{0.5563} \approx 1.74$, moderately increasing the risk compared to the baseline.
- SatisfactionScore5 ($e^{(-1.4080)}$): A score of 5 reduces the odds of churning to about $e^{1.4080} \approx 4.09$ times less than the baseline score.
- Complain ($e^{(14.4887)}$): Customers who have lodged complaints have their odds of churning multiplied by $e^{14.4887} \approx 1.97 \times 10^6$, dramatically increasing the likelihood of churning.
- PointEarned ($e^{(-0.0026)}$): Each additional point earned multiplies the odds of churning by $e^{-0.0026} \approx 0.9974$. Each point slightly decreases the odds of churning by approximately 0.26%.
- $\ln(\text{NumOfProduct})$ ($e^{(-17.4843)}$): The natural logarithm of the number of products held multiplies the odds of churning by $e^{-17.4843}$, which is an extremely small factor, indicating a very significant reduction in churning odds.
- NumOfProduct ($e^{(10.3263)}$): Each additional product multiplies the odds of churning by $e^{10.3263} \approx 30590$. This represents a massive increase in the odds of churning per additional product, suggesting a potentially problematic over-engagement.

Marginal Plots:

For quantitative variables



Based on the marginal model plots, model 4 is a valid model for the data. The marginal model plots for Age, Point.Earned, log(NumOfProducts), NumOfProducts and the Linear Predictor shows that the lines for the nonparametric estimates are very close to the lines for the fitted values.

IsActiveMember and Complain are categorical variables.

Anova Test:

Test Hypothesis:

$H_0: \beta_{\log(\text{NumOfProducts})} = 0$

$H_a: \beta_{\log(\text{NumOfProducts})} \neq 0$

Test Statistic:

Difference in deviance between model 3 and model 4 is 11.558.

P – value: 0.0006747

Since P value is less than alpha at 10% significance level, we reject null hypothesis. Therefore, there is sufficient evidence to say that model 4 is the preferred model at 10% significance level.

Wald test:

The Wald test is a statistical procedure used to assess the significance of coefficients in statistical models by comparing the estimated coefficient to its standard error to determine if it is significantly different from a hypothesized value.

Predictor	P-value
Age	0.000
IsActiveMember	0.011
SatisfactionScore2	0.538
SatisfactionScore3	0.061
SatisfactionScore4	0.611
SatisfactionScore5	0.175
NumOfProducts	0.022
Points.Earned	0.56
Complain	0.00
Log(NumOfProducts)	0.11

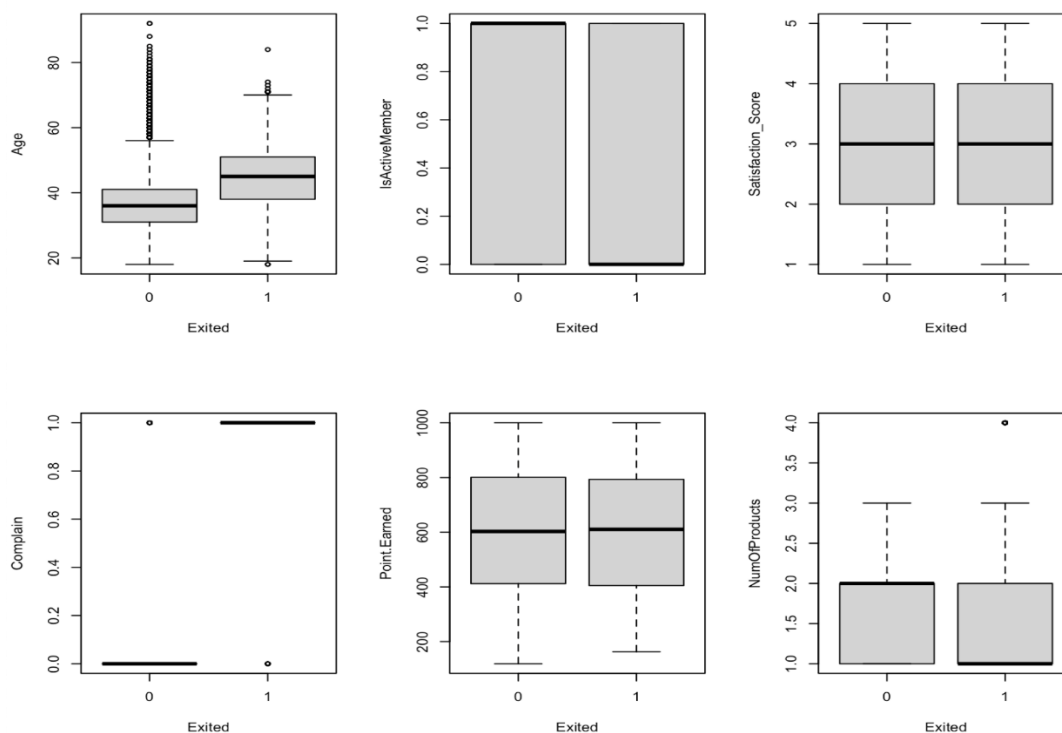
The one highlighted in red signifies that, the predictor Variable is not significant.

ODD's Ratio at 90% Confidence Interval:

Predictor	90% Confidence Interval
Age	(1.044,1.134)
IsActiveMember	(0.0757,0.574)
SatisfactionScore2	(0.317,12.468)
SatisfactionScore3	(0.029,0.796)
SatisfactionScore4	(0.287,10.571)
SatisfactionScore5	(0.044,1.347)
NumOfProducts	(1.812e+01,5.142e+07)
Points.Earned	(0.995,0.999)
Complain	(360462.7,10663592.3)
Log(NumOfProducts)	(2.931e-13,2.218e-03)

The one highlighted in red signifies that, the predictor Variable is not significant because it contains 1 in the interval.

BOXPLOT:



1. Age:

- Group '0' has a younger median age than Group '1'.
- The age range (interquartile range, IQR) is smaller for Group '0' compared to Group '1'.
- There are outliers in both groups, indicating some very young or very old individuals that are exceptions to the general age distribution.

2. IsActiveMember:

- This variable is likely categorical or binary (0 or 1), represented by the height of the boxes, which appear to be constant and likely represent proportions.
- The plot suggests there might not be a significant difference in the proportions of active members between the two groups.

3. Satisfaction_Score:

- Both groups have similar median satisfaction scores, but Group '1' has a slightly wider IQR, suggesting more variability in satisfaction.
- There are outliers in both groups, particularly Group '0', which suggests some members have unusually low or high satisfaction scores.

4. Complain:

- Like IsActiveMember, this appears to be a binary variable, with the boxes representing the proportion of complaints.
- Group '0' shows a lower proportion of complaints (nearly zero), while Group '1' has a noticeably higher proportion of complaints.

5. Point_Earned:

- There's a noticeable difference in median points earned between the two groups, with Group '0' earning more points on average.
- Group '1' has a wider IQR and more outliers, indicating greater variability in the points earned.

6. NumOfProducts:

- The median number of products is similar between the two groups, though Group '1' shows a slightly higher median.
- Both groups have outliers, particularly Group '1', indicating some members use a significantly higher number of products than the rest.

Multicollinearity:

We check for multicollinearity to ensure that predictor variables are not highly correlated, which can destabilize the regression model, inflate standard errors, and make it difficult to determine the individual effect of each predictor.

Predictor	90% Confidence Interval
Age	1.11
IsActiveMember	1.09
SatisfactionScore	1.09
NumOfProducts	1.71
Points.Earned	1.12
Complain	1.02

Hosmer-Lemeshow Test:

Test Hypothesis:

H0: Logistic regression model provides adequate fit to the data.

Ha: The model does not adequately fit the data.

Number of Groups :

Total number of predictors + 1 = 6+1 = 7

Total grouping must be greater than 7, therefore we choose group = 8

Test Statistics:

Chi-squared value:2.9056

Degrees of Freedom (DF): 6

P-value: 0.8206

We fail to reject H_0 , therefore we have insufficient evidence at 5% significance level to conclude that our model does not adequately fit the data.

Model Comparison using AIC/BIC:

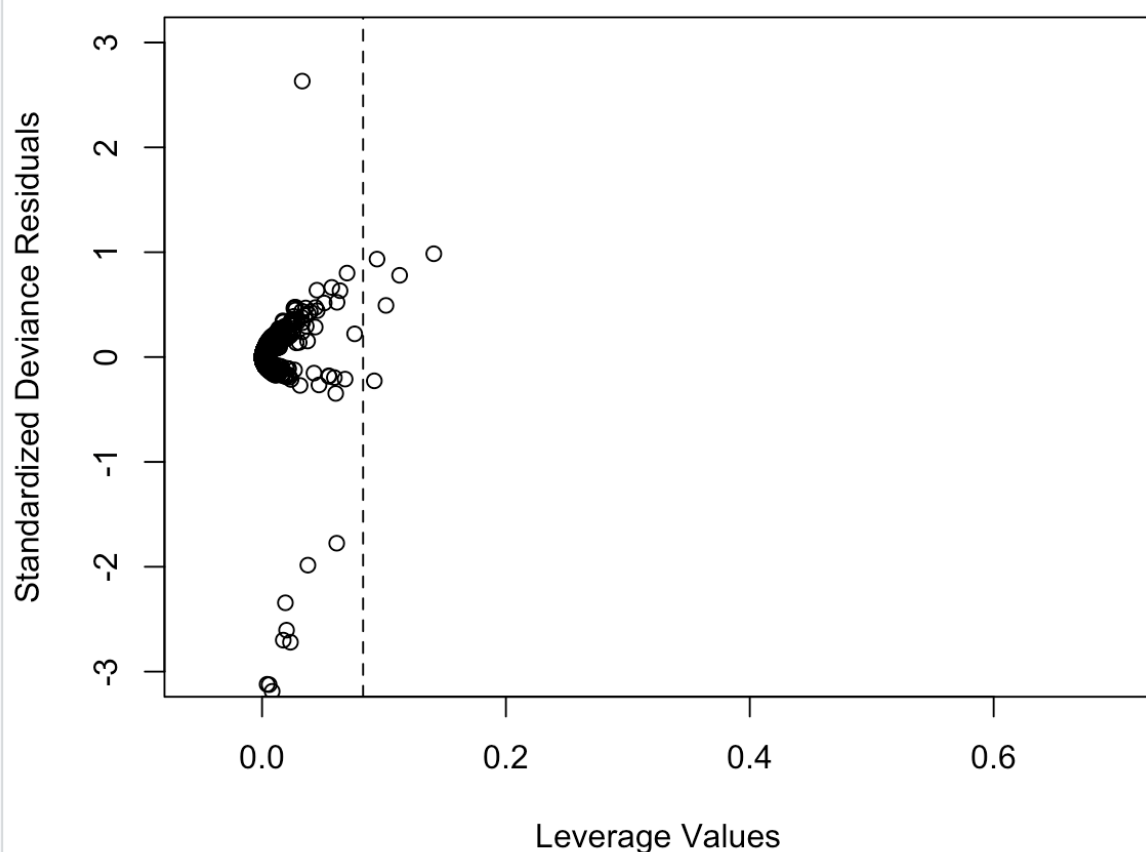
We know that lower the AIC/BIC value better the model,

MODEL 3: $\text{Exited} \sim \text{Age} + \text{IsActiveMember} + \text{Satisfaction_Score} + \text{Complain} + \text{Point.Earned} + \text{Balance} + \text{NumOfProducts}$

MODEL 4: $\text{Exited} \sim \text{Age} + \text{IsActiveMember} + \text{Satisfaction_Score} + \text{Complain} + \text{Point.Earned} + \log(\text{NumOfProducts}) + \text{NumOfProducts}$

	Model 3	Model 4
AIC	180	170
BIC	252	249

Leverage v/s Residual Plot:



- Ideally, points should be randomly scattered, without any apparent pattern. This would indicate that the variance of the residuals is constant (homoscedasticity) and that the model fits the data well.
- In this plot, we see a concentration of points around the lower leverage values, which is typical, as most observations are not influential. There is also a cluster of points at the high leverage end, indicating some observations with a potentially high influence on the model's estimates.

Conclusion:

- Best Model that we found has Predictors: Age, IsActiveMember, Satisfaction_Score, Complain, Points.Earned, NumOfProducts.
- We carefully compared their AIC & BIC value to check which model is better, while we also used Anova.
- After careful calculations Age and Feedback is considered to be the most crucial factor.

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